Orthogonal Rotation of Principal Component Factors in Affine Term Structure Models

Jan Alsterlind

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Abstract

This paper extends the affine term structure model (ATSM) of Adrian, Crump, and Moench (2013) (ACM) by introducing a rotation and scaling procedure that yields economically identifiable pricing factors. More generally, the proposed methodology applies to any factor-based term structure model with observed factors obtained from principal component analysis (PCA). While PCA factors successfully capture the variation in the yield curve, their economic interpretation is limited because principal components are only identified up to an orthogonal rotation. As a result, the incorporation of external economic information into factor dynamics is not straightforward. To address this issue, we apply a sequence of Givens rotations that impose economically motivated restrictions on the factor loading matrix. The first factor is subsequently rescaled and aligned with the short-term risk-free interest rate, while the remaining factors are constructed to be orthogonal to the short end of the yield curve. This transformation preserves the statistical information contained in the original factor representation while producing factors with a direct economic interpretation. The resulting factor structure provides a natural framework for incorporating external information into the estimation of factor dynamics. In particular, it facilitates the incorporation of long-run beliefs about equilibrium short-term interest rates through the hierarchical prior framework of Villani (2009), as well as survey-based expectations of future interest rates in the spirit of Kim and Orphanides (2005). By linking yield curve factors to economically observable quantities, the proposed approach improves identification, supports the integration of macroeconomic information, and preserves the tractability of the ACM framework and other observed-factor term structure models for estimation and forecasting.

1 Orthogonal Rotation of Principal Component Factors in Affine Term Structure Models

This paper presents an extension of the yield curve modeling framework introduced by Adrian, Crump, and Moench (2013), hereafter ACM. The ACM model extracts latent pricing factors from a panel of zero-coupon bond yields using principal component analysis (PCA) and models their dynamics through a vector autoregression (VAR). Although highly successful in explaining the cross-sectional and time-series variation of the yield curve, the factor representation remains largely statistical. Since principal components are identified only up to an orthogonal rotation, the estimated factors lack a unique economic interpretation. This identification problem becomes particularly important when external information is introduced into the model. Researchers often wish to incorporate economically motivated beliefs regarding future short-term interest rates, monetary policy, or long-run equilibrium interest rates. Examples include long-run priors on the steady-state level of interest rates, as in Villani (2009), and survey expectations of future policy rates, as in Kim and Orphanides (2005). However, such information is difficult to incorporate directly into the original ACM framework because the

PCA factors do not correspond to clearly identifiable economic quantities. To address this limitation, we propose a modification that imposes an economically motivated structure on the latent factors through a sequence of Givens rotations and a subsequent scaling procedure. The rotations are chosen such that the first factor becomes uniquely associated with the short end of the yield curve, while the remaining factors are constructed to be orthogonal to it. The first factor is then rescaled to align with the observed short-term risk-free interest rate. The resulting factor representation preserves the statistical properties of the original PCA decomposition but provides a clear economic interpretation of the factors. The main contribution of the paper is therefore not merely improved interpretability. Rather, the rotation creates an economically identifiable factor structure that allows external information to be incorporated directly into the model. Long-run beliefs can be imposed on the unconditional mean of the short-rate factor within a Bayesian VAR framework, while survey expectations can be incorporated as restrictions on the expected path of future interest rates. In this way, the proposed approach provides a bridge between the statistical factor structure of the ACM model and economically observable measures of expectations. The remainder of the paper proceeds as follows. First, we review the ACM framework and the relationship between PCA and latent factor models. We then introduce the Givens rotation methodology and show how economically motivated restrictions can be imposed on the loading matrix while preserving the statistical content of the factor space. Finally, we demonstrate how the rotated factors facilitate the incorporation of long-run beliefs and survey expectations within a Bayesian VAR framework for yield curve analysis.

1.1 The ACM Yield Curve Model

The model developed by Adrian, Crump, and Moench (2013) is part of the class of affine term structure models (ATSM), which describe bond yields as affine functions, that is, linear combinations plus a constant, of the underlying k factors. This structure provides analytical tractability while allowing for a flexible representation of the yield curve as described by Duffie and Kan (1996). In this framework, the yield on a zero-coupon bond with maturity n at time t is modeled as:

$$y_t(n) = A(n) + B(n)^\top F_t + \varepsilon_t(n) \tag{1}$$

where $y_t(n)$ is the observed yield at time t for maturity n, $A(n) \in \mathbb{R}$ is a maturity-specific intercept, $B(n) \in \mathbb{R}^k$ is a vector of maturity-specific factor loadings, $F_t \in \mathbb{R}^k$ is a vector of pricing factors, and $\varepsilon_t(n)$ is an error term assumed to be independently and identically distributed (i.i.d.) with zero mean and constant variance. In the affine no-arbitrage framework, the coefficients $A(n)$ and $B(n)$ are not estimated independently for each maturity. Instead, they are linked across maturities through the exponential-affine bond-pricing equations implied by the no-arbitrage restrictions of the model. Consequently, $A(n)$ and $B(n)$ are determined by the risk-neutral dynamics of the factors.

Following Adrian, Crump, and Moench (2013), the pricing factors are obtained from the cross-section of observed yields and are assumed to follow a first-order vector autoregressive process (VAR(1)):

$$F_{t+1} = \Phi F_t + u_{t+1}, \quad u_{t+1} \sim \mathcal{N}(0, \Sigma) \tag{2}$$

The ACM estimation procedure proceeds in two stages. In the first stage, the factor structure is extracted from the cross-section of yields using principal component analysis (PCA). The resulting principal components are used as pricing factors. In the second stage, the market prices of risk are estimated through cross-sectional regressions that relate bond yields to the estimated factors F_t and their conditional expectations, $\mathbb{E}_t[F_{t+1}]$. These conditional expectations are obtained from the VAR representation of the factor dynamics:

$$\mathbb{E}_t[F_{t+1}] = \Phi F_t \tag{3}$$

Using these estimates, the term premium for a bond with maturity n is defined as the difference between the actual yield and the corresponding risk-neutral yield:

$$\mathrm{TP}_t(n) = y_t(n) - y_t^{\mathrm{RN}}(n) \tag{4}$$

The risk-neutral yield $y_t^{\mathrm{RN}}(n)$ is calculated based on expected future factors but excludes compensation for risk, reflecting the yield that would prevail in a hypothetical risk-neutral world.

1.2 Principal Components and Factor Models

In the ACM model, the latent pricing factors are estimated using principal component analysis (PCA) applied to a panel of zero-coupon bond yields across maturities. As shown by Stock and Watson (2002), principal components can consistently estimate the space spanned by latent factors under mild regularity conditions, especially when the cross-sectional dimension m grows large relative to the time dimension T. A general factor model can be formulated as:

$$Y = F\Lambda + W, \tag{5}$$

where $F \in \mathbb{R}^{T\times k}$ contains the first k principal components (scores), $\Lambda \in \mathbb{R}^{k\times m}$ contains the corresponding loading vectors (eigenvectors), $k \ll m$ is the number of retained components, W is an error vector capturing the approximation error.

Unlike general factor models, PCA assumes orthogonality of components and does not explicitly model the noise structure W but offers a nonparametric alternative to traditional factor estimation, avoiding strong distributional assumptions. However, a key limitation of PCA is the non-uniqueness of its components. While the principal component space is uniquely defined, the individual components are only identified up to an orthogonal rotation. That is, if F denotes the estimated factors, then any rotated version, $\hat{F} = FQ$, where Q is an orthogonal matrix[1], spans the same space and yields identical fit. This rotational indeterminacy implies that interpretability of components depends on post-estimation choices. Such transformations preserve the statistical properties of the estimated factors while aligning them with theoretical or empirical priors. In summary, while PCA and factor models are tightly linked in their ability to recover latent structures, the interpretability of principal components hinges on rotational choices, with methods like Givens rotations offering a principled path toward meaningful representation.

1.3 Reconfiguring Factor Loadings via Givens Rotations

We can alter the interpretation of PCA factors while preserving their statistical properties by a method introduced by Givens (1958), in which a *orthonormal matrix*[2] is used to perform a rotation in the plane spanned by two coordinate axes, typically denoted i and j, in $\mathbb{R}^k$.

[1] An orthogonal matrix is a square matrix whose columns form an orthonormal basis, meaning that the columns have unit length and are mutually orthogonal. Equivalently, $Q^\top Q = QQ^\top = I$, implying $Q^{-1} = Q^\top$.

[2] A matrix Q is orthonormal if its columns form an orthonormal set, meaning that each column has unit length and different columns are mutually orthogonal.

The corresponding *Givens rotation matrix*, $G(i, j, \theta)$, is derived from the identity matrix by modifying the entries in rows and columns i and j to implement the rotation:

$$G(i,j,\theta) = \begin{pmatrix} 1 & 0 & \cdots & & & & & \cdots & 0 \\ 0 & 1 & \cdots & & & & & \cdots & 0 \\ \vdots & \vdots & \ddots & & & & & & \vdots \\ & & & \cos(\theta) & \cdots & \sin(\theta) & & & \\ & & & \vdots & \ddots & \vdots & & & \\ & & & -\sin(\theta) & \cdots & \cos(\theta) & & & \\ \vdots & \vdots & & & & & \ddots & & \vdots \\ 0 & 0 & \cdots & & & & & 1 & 0 \\ 0 & 0 & \cdots & & & & & 0 & 1 \end{pmatrix} \tag{6}$$

The Givens rotation modifies exclusively the i-th and j-th rows and columns, while all remaining diagonal entries are equal to 1 and all off-diagonal entries are zero. The rotation matrix $G(i, j, \theta)$, where θ denotes a rotation angle measured in *radians*[3], mixes columns i and j of Λ, and simultaneously rows i and j of F, preserving orthogonality and vector norms in both matrices. Specifically, the rotated factor matrix is given by a unique transformation based on Givens rotations:

$$\hat{F} = FG(\theta_k)^\top G(\theta_{k-1})^\top \cdots G(\theta_1)^\top \tag{7}$$

$$\hat{\Lambda} = G(\theta_1)G(\theta_2) \cdots G(\theta_k)\Lambda \tag{8}$$

so that the rotated model

$$Y = \hat{F}\hat{\Lambda} + W \tag{9}$$

remains equivalent to the original. These rotations affect only the interpretation of the factors and their loadings, not the statistical content of the model. Gürkaynak, Sack, and Swanson (2005) apply a rotation technique to disentangle monetary policy surprises—measured by changes in federal funds futures across different maturities—into a target factor and a path factor.

1.4 Sparse Representation of Yield Curve Loadings through Givens-Based Factor Rotation

Let $\Lambda \in \mathbb{R}^{k \times m}$ denote the loading matrix obtained from PCA, where m is the number of maturities and k is the number of retained principal components. Let $F \in \mathbb{R}^{T \times k}$ denote the corresponding factor time series. Since PCA factors are only identified up to an orthogonal rotation, we exploit this rotational freedom to obtain an economically meaningful factor representation. Specifically, we apply a sequence of Givens rotations to construct a rotated loading matrix $\hat{\Lambda}$ satisfying

$$\hat{\lambda}_{i1} = 0, \qquad i = 2, \ldots, k. \tag{10}$$

This restriction implies that only the first rotated factor loads on the shortest maturity, while all remaining factors are orthogonalized with respect to the short end of the yield curve. The resulting factor decomposition therefore isolates a short-rate factor and assigns the remaining

[3] A radian is the standard unit for measuring angles. One radian corresponds to the angle subtended by an arc whose length equals the radius of the circle. A full rotation equals 2π radians (360°).

variation to factors associated with medium- and long-term maturities. To impose these restrictions, we construct a sequence of $k-1$ Givens rotations,

$$G(\theta_1), G(\theta_2), \ldots, G(\theta_{k-1}),$$

which are applied iteratively to the loading matrix. For each rotation step $s = 1, \ldots, k-1$, the rotation angle is obtained by solving

$$\theta_s = \arg\min_{\theta} \left| [G(i,j,\theta)\Lambda]_{i1} \right|. \tag{11}$$

For the selected pair of factors, the optimization searches for the angle that minimizes the corresponding loading on the shortest maturity. In practice, the resulting minimum is numerically indistinguishable from zero, implying that the targeted loading is effectively eliminated. When $k = 2$, the solution is available in closed form,

$$\theta^* = \operatorname{arccot}\left(\frac{\lambda_{11}}{\lambda_{21}}\right),$$

where λ_{11} and λ_{21} denote the elements of the first column of Λ. This corresponds to the identification proposed by Gürkaynak, Sack, and Swanson (2005).[4] For higher-dimensional systems ($k > 2$), a numerical solution is more convenient. After applying the full sequence of rotations, the first column of $\hat{\Lambda}$ contains zeros in all rows except the first, thereby satisfying the imposed sparsity constraint.

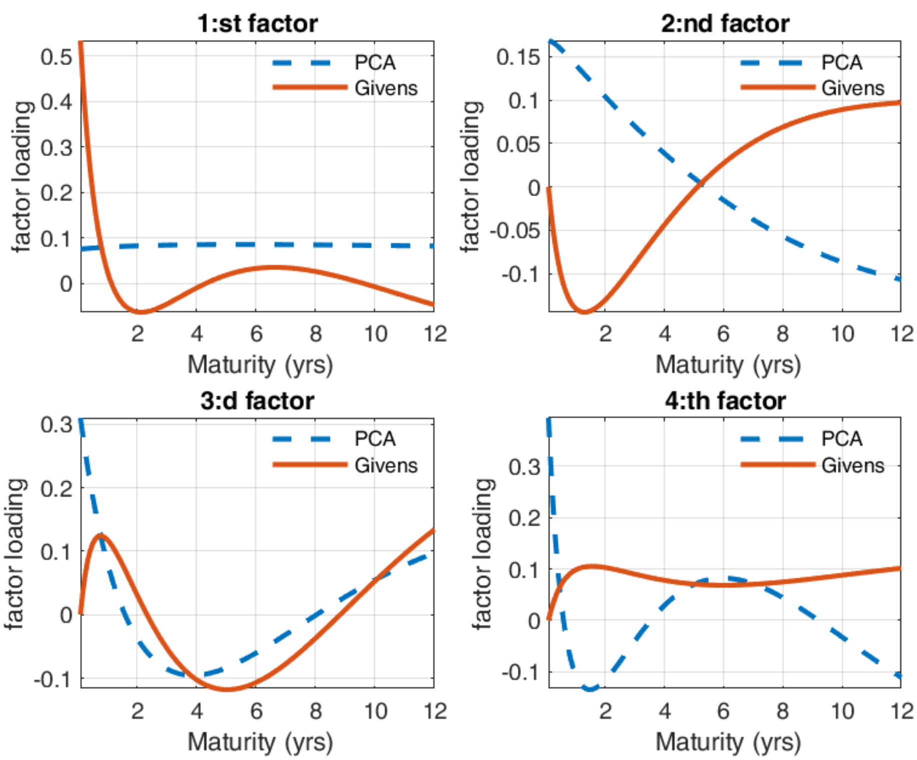

Figure 1: Factor Loadings Across Maturity for PCA and Givens-Rotated Factors, Four-Factor Specification

[4]The restriction follows from $\cos(\theta)\lambda_{21} - \sin(\theta)\lambda_{11} = 0$, implying $\cot(\theta) = \lambda_{11}/\lambda_{21}$.

Figure 1 compares the loading profiles obtained from PCA and from the Givens-rotated factor representation. The first rotated factor is the only factor loading on the shortest maturity, while the remaining factors exhibit zero loadings at the shortest maturity by construction. At the same time, these factors continue to explain variation across medium- and longer-term maturities. The rotation therefore preserves the explanatory power of the original PCA solution while yielding a factor structure that is more closely aligned with economic interpretations of the term structure. The resulting sparsity pattern resembles the objective of the varimax rotation introduced by Kaiser (1958). Varimax seeks a simple loading structure by maximizing the variance of squared loadings within each factor, thereby improving interpretability. Unlike varimax and related interpretability-oriented rotations, however, the Givens rotations considered here serve primarily as identification devices. Their purpose is not to maximize sparsity globally, but rather to impose economically motivated restrictions that uniquely identify a factor associated with the short end of the yield curve.

1.5 Block-Orthonormal Rotations in $\mathbb{R}^k$

The sparsity restriction imposed on the first factor ensures that the rotated loading matrix $\hat{\Lambda}$ isolates a short-term component of the yield curve. As illustrated in Figure 1, the first rotated factor loads strongly on the shortest maturity, while the remaining factors exhibit loadings that are identically zero at the short end. This pattern reflects the effect of the initial sequence of Givens rotations: by eliminating all entries $\hat{\lambda}_{i1}$ for $i > 1$, the first factor becomes anchored at the short end of the maturity spectrum, and the remaining factors are orthogonal to it in their loading structure. Having imposed this restriction, any further rotation of the remaining factors must preserve the structure of the first column. In other words, subsequent transformations must leave the short-end factor unchanged while allowing additional economically meaningful restrictions to be introduced for factors $2, \ldots, k$. This motivates the use of *block-orthonormal rotations*, which operate exclusively on the subspace spanned by the remaining factors and therefore maintain the sparsity pattern established in the first step. Formally, after imposing the restriction

$$\hat{\lambda}_{i1} = 0 \qquad \forall i > 1,$$

the rotated loading matrix takes the form

$$\hat{\Lambda} = \begin{pmatrix} \hat{\lambda}_{11} & \hat{\lambda}_{12} & \cdots & \hat{\lambda}_{1m} \\ 0 & \hat{\lambda}_{22} & \cdots & \hat{\lambda}_{2m} \\ 0 & \hat{\lambda}_{32} & \cdots & \hat{\lambda}_{3m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \hat{\lambda}_{k2} & \cdots & \hat{\lambda}_{km} \end{pmatrix}$$

To preserve this structure, subsequent rotations must act only on the subspace spanned by factors $2, \ldots, k$. We therefore consider orthonormal transformations $Q \in O(k)$ of the block-diagonal form

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & Q_{(k-1)} \end{pmatrix},$$

where $Q_{(k-1)} \in O(k-1)$ is an arbitrary orthonormal matrix operating on the reduced factor space. The transformed loading matrix is given by

$$\tilde{\Lambda} = Q\hat{\Lambda}.$$

Because the upper-left element of Q is fixed at unity and the remaining entries in the first row and column are zero, the first column of $\hat{\Lambda}$ is left invariant:

$$\tilde{\lambda}_{i1} = \hat{\lambda}_{i1} \qquad \forall i.$$

Thus, the short-end restriction imposed in the first rotation is preserved exactly. A natural second identifying restriction is to construct a factor that is anchored at the long end of the yield curve. Having identified the first factor, we restrict all subsequent rotations to the subspace spanned by factors $2, \ldots, k$. We parameterize the remaining orthonormal transformation as a sequence of Givens rotations,

$$Q_{(k-1)}(\theta) = G_1(\theta_1)G_2(\theta_2)\cdots G_r(\theta_r),$$

where each rotation acts only within the reduced $(k-1)$-dimensional factor space. The corresponding block-orthonormal transformation is

$$Q(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & Q_{(k-1)}(\theta) \end{pmatrix}.$$

The rotated loading matrix becomes

$$\tilde{\Lambda} = Q(\theta)\hat{\Lambda}.$$

For each rotation step s, the angle θ_s is chosen to maximize the loading of the second factor at the longest maturity:

$$\theta_s = \arg\max_{\theta} \left| \left[G(i,j,\theta)\hat{\Lambda} \right]_{2m} \right|. \tag{12}$$

Here, $(2, m)$ denotes the loading of the second factor at maturity m. The optimization is performed numerically, and the resulting sequence of Givens rotations defines the block-orthonormal transformation $Q_{(k-1)}$. Because all rotations are restricted to the subspace spanned by factors $2, \ldots, k$, the short-end restriction imposed on the first factor remains unchanged. The resulting second factor is therefore aligned as closely as possible with the longest maturity, conditional on the identification of the first factor. When the remaining factor space is two-dimensional, only a single rotation angle is required. In higher-dimensional settings, the same principle applies, although multiple Givens rotations are needed to parameterize the admissible block-orthonormal transformations. The long-end restriction represents only one possible identifying assumption. More generally, the remaining rotational degrees of freedom may be used to impose additional economically motivated restrictions, such as concentrating factor loadings at selected maturities, constructing factors associated with slope or curvature movements, or incorporating information from external sources. Because all such transformations are restricted to the subspace spanned by factors $2, \ldots, k$, the short-end identification of the first factor is preserved throughout.

1.6 Invariant Transformations and Economic Identification of Yield Curve Factors

A key contribution of Joslin, Singleton, and Zhu (2011) is the development of an affine term structure model formulated directly in terms of observed yield curve factors rather than latent state variables. Building on the widespread use of principal components as empirical representations of yield curve dynamics, they demonstrate that affine transformations of the factor space leave model-implied bond yields unchanged. More specifically, if the factors are transformed according to

$$F_t^* = DF_t + C, \tag{13}$$

where D is an invertible matrix and C is a constant vector, the remaining model parameters can be adjusted so that the yield curve implied by the model is unchanged. Consequently, multiple factor representations may generate identical bond prices and yields, implying that additional identifying restrictions are required to obtain an economically meaningful factorization. This invariance property is particularly useful when working with observed yield curve factors extracted from principal component analysis. Since the economic interpretation of principal components is generally not unique, affine transformations can be used to obtain factors with a clearer economic meaning without affecting the model's pricing implications. In this paper, the extracted factors are first subjected to a sequence of Givens rotations designed to produce a sparse loading structure in which the first factor loads primarily on the shortest maturity. This rotation aims to isolate the component of the yield curve most closely related to short-rate dynamics. This rotation aims to isolate the component of the yield curve most closely related to short-rate dynamics. The resulting factor is then rescaled to match the observed short-term interest rate. Let $f_{1,t}^{\text{rot}}$ denote the rotated first factor and let y_{1t} denote the observed short-term interest rate. The rescaling is obtained from the regression

$$y_{1t} = \alpha + \beta f_{1,t}^{\text{rot}} + \varepsilon_t, \tag{14}$$

yielding the transformed factor

$$f_{1,t}^{\text{scaled}} = \hat{\alpha} + \hat{\beta} f_{1,t}^{\text{rot}}. \tag{15}$$

This transformation is a special case of the affine invariant transformations studied by Joslin, Singleton, and Zhu (2011). Because it only changes the representation of the factor space, the model-implied yield curve and all pricing results remain unchanged. At the same time, the transformation provides an economically interpretable identification in which the first factor can be viewed as a short-rate factor expressed directly in the units of the observed short-term interest rate. The identification procedure proposed here is not specific to the ACM model. Rather, it can be applied to any affine term structure model based on observed yield curve factors, exploiting the affine invariance property emphasized by Joslin, Singleton, and Zhu (2011) to obtain economically interpretable factor representations without affecting bond-pricing implications.

Figure 2 illustrates the effectiveness of the identification procedure by comparing the observed short-term interest rate (blue line) with the rotated and rescaled first factor extracted from principal component analysis (red line). Each subplot corresponds to a specification with between two and five factors. As the dimensionality of the factor space increases, the first factor increasingly resembles the observed short-term rate, suggesting that higher-dimensional factor structures facilitate a cleaner separation of short-rate dynamics from other yield-curve movements. The improvement is most pronounced when moving from two to four factors, while the gains beyond four factors appear relatively modest. The five-factor specification employed by Adrian, Crump, and Moench (2013) delivers the closest correspondence between the identified factor and the observed short-term rate.

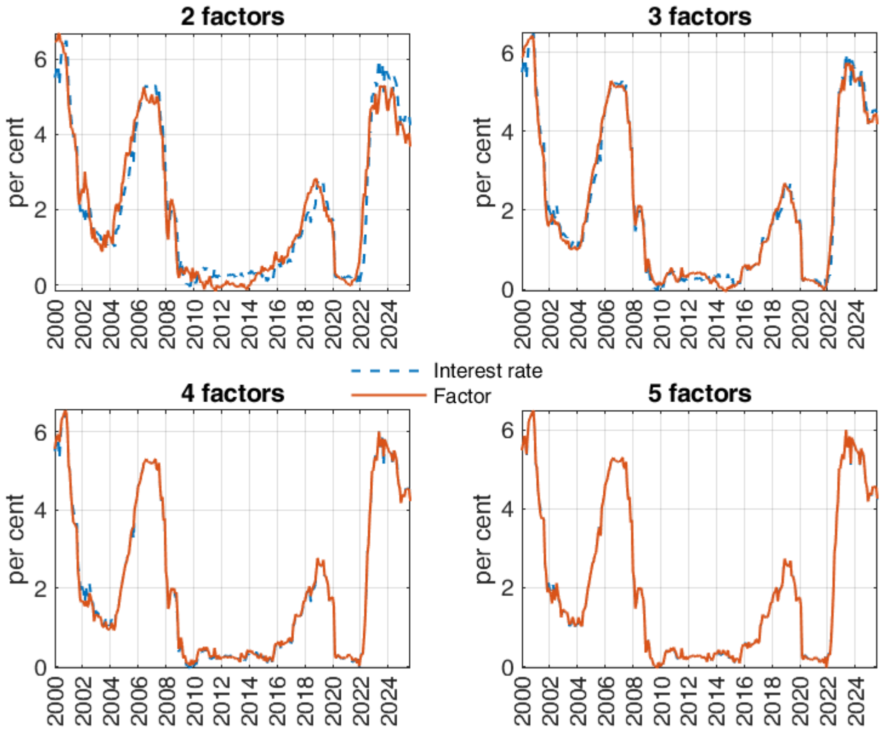

Figure 2: Short-Term Risk-Free Interest Rate and Scaled Givens-Rotated First Factor Across Specifications with two, three, four and five Factors

2 Bayesian VAR Estimation with Long-Run Priors and Survey Expectations

The rotation and scaling procedure introduced in the previous section serves a purpose beyond improving factor interpretability. By aligning the first factor with the short-term interest rate and expressing the remaining factors in economically meaningful dimensions, the transformed factor representation provides a natural framework for incorporating external information about the future path of interest rates. In the original ACM specification, the latent factors are principal components whose orientation is statistically determined and therefore difficult to relate directly to economic beliefs. As a consequence, imposing prior information on long-run interest rate expectations or survey forecasts is not straightforward. Following the Givens rotations and the subsequent scaling procedure proposed in this paper, the factors acquire a clear economic interpretation, allowing external information to be incorporated in a transparent and economically meaningful manner. Two complementary sources of information are considered. First, following Villani (2009), long-run beliefs are introduced through hierarchical priors on the unconditional mean of the VAR. These priors provide information about the steady-state levels of the pricing factors and are particularly useful for anchoring the long-run level of the short-rate factor. Second, survey expectations, in the spirit of Kim and Orphanides (2005), are incorporated as noisy observations of model-implied forecasts. Unlike long-run priors, survey information primarily informs the expected transition dynamics of the factors over specific forecast horizons. Together, these approaches allow economically motivated information to influence both the long-run behavior and the short- to medium-term dynamics of the yield curve. To model the dynamics of the rotated factors, we employ the

Bayesian VAR framework of Villani (2009). Let

$$F_t^* = (f_{1,t}^{\text{rotated, scaled}}, f_{2,t}^{\text{rotated}}, \dots, f_{n,t}^{\text{rotated}})^\top$$

denote the vector of rotated and transformed pricing factors extracted from the yield curve. The BVAR(1) specification is

$$F_t^* = \mu + \Phi(F_{t-1}^* - \mu) + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \Sigma), \tag{16}$$

where μ is the vector of unconditional means, Φ is the autoregressive coefficient matrix, and Σ is the innovation covariance matrix. The mean-centered representation provides a convenient environment for incorporating both long-run priors and survey-based restrictions, since the rotated first factor can be interpreted as a proxy for the short-term interest rate and its expected future path.

2.1 Incorporating Long-Run Beliefs via Hierarchical Priors

Villani (2009) proposes a hierarchical prior structure that explicitly targets the unconditional mean of the VAR process. Specifically, the unconditional mean vector, μ, is treated as an unknown parameter and assigned its own prior distribution. This framework enables the incorporation of economically meaningful beliefs about long-run levels, such as the equilibrium short-term interest rate, while allowing the remaining model parameters to be estimated from the data. The hierarchical setup is particularly attractive in yield curve applications, where anchoring certain factors to plausible long-run values can improve the economic interpretation of the model. The prior distribution of the unconditional mean vector is specified as

$$\mu \sim \mathcal{N}(\mu_0, V_\mu), \tag{17}$$

where μ_0 denotes the prior mean vector and V_μ the prior covariance matrix. The posterior distribution of μ is updated jointly with the VAR parameters using Gibbs sampling. In this application, the prior mean vector is specified as

$$\mu_0 = \begin{bmatrix} y^* & 0 & \cdots & 0 \end{bmatrix}', \tag{18}$$

and the prior covariance matrix as

$$V_\mu = \operatorname{diag}\left(\sigma_{\mu,1}^2, \sigma_{\mu,2}^2, \dots, \sigma_{\mu,n}^2\right). \tag{19}$$

The parameter y^* represents the subjective assessment of the long-run equilibrium short-term interest rate and reflects beliefs regarding the long-run stance of monetary policy. The first factor is therefore anchored to an economically meaningful steady-state level, while the remaining factors are assigned prior means of zero. This is consistent with their interpretation as principal components extracted from the yield curve, which are normalized to have zero sample means. The diagonal covariance matrix allows different degrees of uncertainty to be assigned to each element of μ. In particular, $\sigma_{\mu,1}^2$ determines the strength of the prior anchoring of the short-rate factor, whereas the remaining variances govern the degree of shrinkage toward zero for the other factors. Consequently, the specification incorporates economically motivated long-run beliefs while preserving flexibility in the estimation of the remaining dynamics, in the spirit of Villani (2009). To illustrate the role of long-run beliefs, the model is estimated using alternative specifications for the prior mean of the short-rate factor. Specifically, y^* is set to 2.5, 3.0, and 4.5 percent, while the prior variances are chosen to reflect a moderate degree of uncertainty around the assumed steady-state values. The estimated factors F_t^* are subsequently used in the representation of the affine yield curve:

$$y_t(n) = A(n) + B(n)^\top F_t^* + \eta_t(n) \tag{20}$$

where $y_t(n)$ denotes the yield at time t with maturity n, and $A(n)$ and $B(n)$ are maturity-specific intercept and loading vectors. The error term $\eta_t(n)$ captures the measurement noise.

This modeling approach is consistent with the empirical strategy employed in the ACM framework and offers a coherent structure for extracting and forecasting latent yield curve factors while incorporating long-term economic beliefs. As emphasized by Meldrum and Roberts-Sklar (2015), the use of informative priors on unconditional means in affine term structure models can enhance the stability of the estimated dynamics.

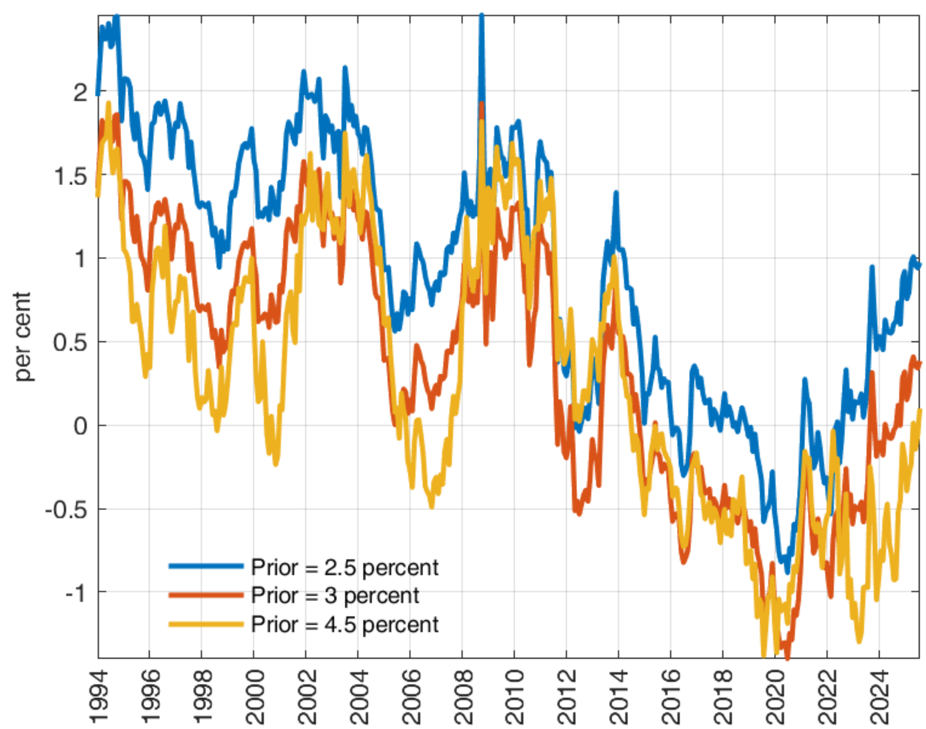

Figure 3: Estimated 5-Year US Term Premium Using ACM and Alternative Priors for the Short-Term Interest Rate. The estimates are based on monthly data from January 1994 to August 2025, using maturities of 1 and 6 months, and 1 to 11 years.

Figure 3 illustrates how the model's estimate of the term premium is shaped by the choice of prior in the framework of Villani (2009). This effect arises because the prior influences the model's forecasts of the short-term interest rate, which in turn feed directly into the calculation of the term premium. When a prior is used that centers around the sample mean of the short rate since 1994—a period characterized by relatively low interest rates—the estimated term premium is notably higher. Conversely, employing a prior centered around the sample mean since 1961, which includes periods of higher average rates, results in a lower estimated term premium. A prior that lies between these two extremes yields an intermediate estimate of the term premium. These results highlight the sensitivity of term structure estimates to prior assumptions, underscoring the importance of carefully selecting priors in empirical applications.

2.2 Incorporating Survey Expectations through Predictive Density Restrictions

The preceding analysis was conducted within the Adrian, Crump and Moench (ACM) framework. To demonstrate that the use of rotated factors is not specific to ACM, this section considers an alternative affine term structure model based on the linear Gaussian framework of Diez de los Rios (2015). The objective is not to advocate a particular model specification or method for incorporating survey information, but rather to illustrate that the economic identification scheme developed above can be implemented in a broad class of term structure models estimated using observed yield curve factors. Consequently, any effects attributed to the factor rotation should be interpreted as arising from the identification of the factors themselves rather than from specific features of the ACM framework. An alternative way of incorporating information about the expected path of short-term interest rates is to use survey expectations rather than direct priors on the unconditional mean of the factors. This approach is closely related to Kim and Orphanides (2005), who augment affine term structure models with survey forecasts of future short rates. Whereas the steady-state prior of Villani (2009) primarily influences the unconditional mean of the VAR, survey expectations provide information about the conditional dynamics of the factors over specific forecast horizons. In the empirical application, the yield curve is represented by three factors extracted from the same panel of Treasury yields used in the baseline ACM specification. Survey information from the Survey of Professional Forecasters (SPF) is incorporated through additional measurement restrictions on expected future short rates, following the spirit of Kim and Orphanides (2005) and the linear Gaussian setting of Diez de los Rios (2015). The objective is not to propose a new or optimal way of incorporating survey expectations into term structure models. Rather, the framework provides a simple and transparent setting in which the effect of survey information on estimated expectations and term premia can be evaluated while maintaining close comparability with the baseline ACM results. The survey data are obtained from the Survey of Professional Forecasters (SPF) and consist of forecasts of the three-month Treasury bill rate at six- and twelve-month horizons. These survey forecasts are interpreted as noisy observations of future short-rate expectations and therefore provide information about the expected evolution of the yield curve beyond that contained in contemporaneous bond prices.

With the rotated-factor specification employed here, the first factor is constructed to closely track the short-term interest rate. Let the first element of F_t^* be denoted by $f_{1,t}$. Under the VAR(1) dynamics

$$F_t^* = \mu + \Phi(F_{t-1}^* - \mu) + \varepsilon_t, \tag{21}$$

the model-implied expectation of the short rate at horizon h is

$$m_t(h) = e_1' \left[(I - \Phi^h)\mu + \Phi^h F_t^* \right], \tag{22}$$

where e_1 is a selection vector with unity in the first position and zeros elsewhere. Suppose survey forecasts are available for horizons $h_1, \ldots, h_K$. Let

$$s_t = \begin{bmatrix} s_t(h_1) \\ \vdots \\ s_t(h_K) \end{bmatrix} \tag{23}$$

denote the corresponding survey observations. Following the spirit of Kim and Orphanides (2005), the discrepancy between survey expectations and model-implied forecasts can be interpreted as a forecast error,

$$u_t = s_t - m_t, \tag{24}$$

where m_t is the vector of model-implied expectations at the same horizons. The survey forecasts are then treated as noisy observations of the model-consistent conditional expectations,

$$u_t \sim \mathcal{N}(0, \Omega), \tag{25}$$

with covariance matrix Ω governing the degree of confidence placed on survey information. Conditional on the VAR parameters, the contribution of the surveys to the likelihood function is therefore

$$L_{\text{survey}} \propto |\Omega|^{-T_s/2} \exp\left(-\frac{1}{2}\sum_{t=1}^{T_s} u_t' \Omega^{-1} u_t\right), \tag{26}$$

where T_s denotes the number of survey observations. In contrast to the Gaussian conditional posteriors obtained in the original Gibbs sampler of Villani (2009), the inclusion of survey information introduces nonlinear dependence on the VAR transition matrix Φ through the terms Φ^h. As a consequence, the conditional posterior distribution of (Φ, μ) is no longer available in closed form. The Gibbs sampler must therefore be augmented with a Metropolis-Hastings step. Specifically, after drawing the covariance matrix Σ and the remaining parameters from their standard conditional posterior distributions, a candidate draw (μ^*, Φ^*) is generated from a proposal density. The acceptance probability is

$$\alpha = \min\left(1, \frac{p(Y|\mu^*, \Phi^*, \Sigma) L_{\text{survey}}(\mu^*, \Phi^*) p(\mu^*, \Phi^*)}{p(Y|\mu, \Phi, \Sigma) L_{\text{survey}}(\mu, \Phi) p(\mu, \Phi)}\right), \tag{27}$$

where $p(Y|\mu, \Phi, \Sigma)$ denotes the standard VAR likelihood and $p(\mu, \Phi)$ represents the prior distribution. The survey likelihood therefore acts as an additional weighting mechanism, favoring parameter configurations that generate forecasts consistent with observed survey expectations.

Figure 4 illustrates the impact of incorporating survey expectations on the estimated term premium. Although the survey-augmented specification is estimated using the same yield data and factor structure as the baseline model, the additional survey restrictions alter the decomposition of yields into expectations and term premia. This suggests that survey expectations contain economically meaningful information about future short rates that is not fully captured by bond yields alone. The survey-based approach can be viewed as a dynamic complement to informative steady-state priors. Whereas long-run mean priors primarily identify the unconditional mean μ, survey expectations provide information about the expected path of future short rates and therefore help identify the persistence parameters in Φ. Because both sources of information influence expectations of future rates, they may interact strongly in estimation. In practice, careful calibration of both the survey error covariance matrix Ω and the prior covariance matrix V_μ is therefore required to avoid excessive identification of the same underlying dynamics.

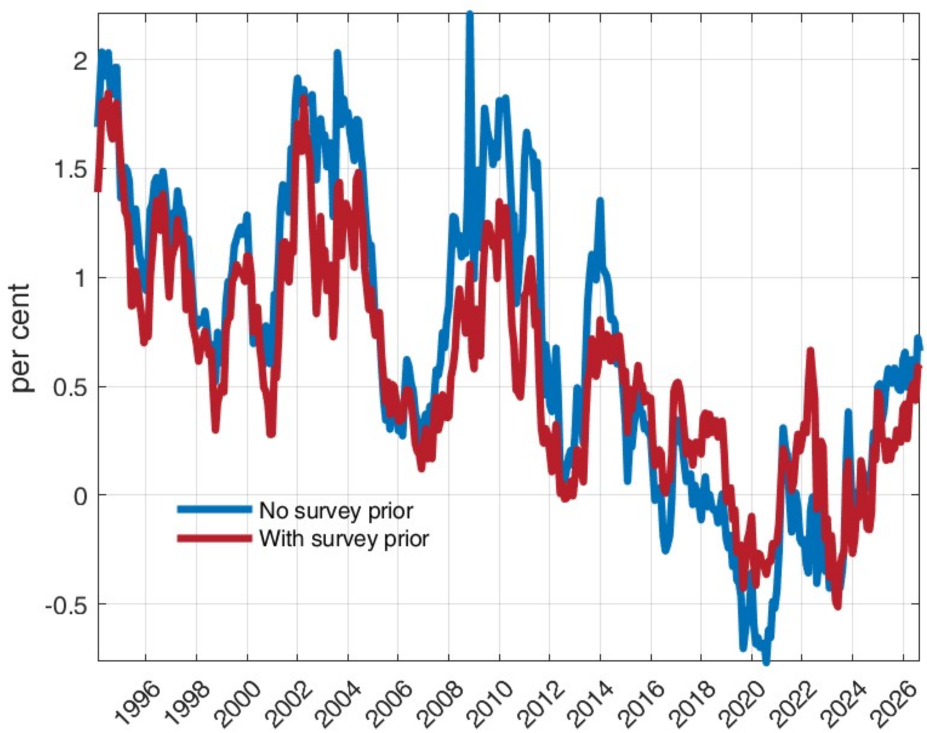

Figure 4: Estimated term premium from the baseline ACM-style model and from the specification augmented with SPF forecasts of the three-month Treasury bill rate at six- and twelve-month horizons.

3 Discussion

The proposed modification preserves the information contained in the PCA factor space while providing a more economically interpretable representation. By imposing sparsity in the loading matrix and aligning the first factor with the short-term interest rate, the model improves the interpretability of yield curve dynamics and facilitates the incorporation of external information into term structure models. Such information may take the form of long-run beliefs about equilibrium interest rates or survey-based expectations regarding the future path of monetary policy. The rotation therefore serves not only as a tool for factor interpretation, but also as a mechanism for linking latent yield curve factors to economically observable quantities.

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